

Data-Driven Social Engagement Initiative

Strategy & Analytics Report

This report implements the architecture described in the supplied project brief: content performance tracking, virality prediction, NLP-based relatability analysis, A/B testing, optimization recommendations, growth visualization, and trend forecasting. The supplied brief describes the system but does not provide live platform credentials or a real dataset; therefore the numerical results below come from a reproducible synthetic dataset.

1. Dataset

The demo contains 5,000 content records and 15,000 comments spanning January–August 2026. Features include platform, topic, format, hook, caption style, video length, impressions, likes, comments, shares, saves, retention, and follower growth.

2. Viral Coefficient

The implementation gives greater weight to high-value actions: shares $\times 3$, saves $\times 2$, comments $\times 1.5$ and likes $\times 0.5$, normalized by impressions. This directly operationalizes the brief's emphasis on shares and saves over passive likes.

3. Top Topics

Topic	Avg Engagement %	Avg Viral Coeff.	Save/Share	Followers Gained
Social Anxiety	9.48	8.703	1.01	179,730
Dating	8.17	7.477	1.03	151,546
Friendship	8.21	7.474	1.03	142,555
Self-Doubt	8.01	7.350	1.01	139,119
Burnout	7.82	7.082	1.09	151,546

4. A/B Testing Findings

Format: Short vs Long. Mean engagement was 8.38% vs 7.09% (-15.4% lift for the second variant); $p=1.18e-49$. Statistically significant at $\alpha=0.05$.

Hook: Visual vs Text. Mean engagement was 8.56% vs 7.18% (-16.1% lift for the second variant); $p=2.826e-59$. Statistically significant at $\alpha=0.05$.

Caption style: Story ranked highest versus List lowest, with an observed difference of 18.5% in mean engagement.

5. NLP / Relatability

The comment module tags comments as Relatable or Neutral using transparent linguistic trigger patterns. On the synthetic benchmark labels, the simple classifier achieved 86.0% agreement. For production, the rule-based approach should be replaced or augmented with a trained transformer/linear NLP classifier and validated on human-labeled comments.

6. Recommended Strategy

Prioritize the strongest-performing topic (Social Anxiety) while continuously testing hooks, formats and caption styles. Use saves and shares as primary quality signals, not likes alone. Use the dashboard to identify content clusters with high save-to-share ratios and retention, then feed those combinations into the weekly recommender.

7. Limitations & Production Path

Synthetic data cannot establish real-world audience behavior. API rate limits, platform-specific metrics, changing algorithms, selection bias, missing comments and confounding variables must be addressed in production. The next step is authenticated API ingestion, human-labeled NLP training data, scheduled model retraining, experiment assignment, and deployment behind secure secrets management.